



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)
ALLEN COMMON TEST
31-10-2021

JEE(Main + Advanced) : NURTURE COURSE

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C,D	A,B	A,C,D	A,D	A,C	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	32.00	5.87	100.00	9.80	0.00	16.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	2	8	6	8	8

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,D	B,C	A,B	A,D	A,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	27.24	61.00	7.00	1.33	93.16	26.18
SECTION-III	Q.	1	2	3	4	5	6
	A.	9	6	1	4	0	7

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,C	A,B,C,D	A,B,C,D	A,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	5.00	2.00	4.33	5.00	0.56	8.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	2	4	1	7	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (B,C,D)**

COM will be on the line $y = x$

2. **Ans (A,B)**

$$\vec{F} = y \hat{i} - x \hat{j}$$

$$W = \int y dx - \int x dy$$

For OA

$$y = dy = 0$$

$$\Rightarrow W = 0$$

For AB

$$x = a, dx = 0, y \text{ from } 0 \text{ to } a$$

$$W = 0 - \int_0^a a dy = -a^2$$

For BC

$$y = a, dy = 0, x \text{ from } a \text{ to } 0$$

$$W = \int_a^0 a dx = -a^2$$

For CD

$$x = d, dx = 0$$

$$W = 0$$

3. **Ans (A,C,D)**

$\hat{s}$ = Unit vector in the direction of displacement

$$= \frac{3\hat{i} - 4\hat{j} - 5\hat{k}}{5\sqrt{2}}$$

$$\text{Initial velocity vector} = \vec{v} = 10\sqrt{2} \left(\frac{3\hat{i} - 4\hat{j} - 5\hat{k}}{5\sqrt{2}} \right)$$

$$= (6\hat{i} - 8\hat{j} - 10\hat{k}) \text{ m/s}$$

$$W = \vec{F} \cdot \vec{s} = (3\hat{i} + 4\hat{j}) \cdot (3\hat{i} - 4\hat{j} - 5\hat{k}) = -7J$$

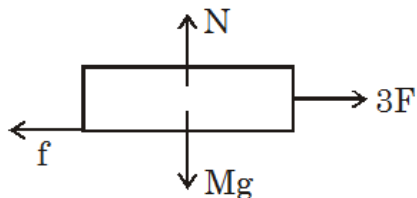
$$P = \vec{F} \cdot \vec{v} = (3\hat{i} + 4\hat{j}) \cdot (6\hat{i} - 8\hat{j} - 10\hat{k}) = -14 \text{ watt}$$

If velocity doesn't change $\Rightarrow F_{\text{net}} = 0 \Rightarrow$ some other force must be present.

4. **Ans (A,D)**

Acceleration of block is in horizontal plane and only friction force is the force able to provide this acceleration.

5. **Ans (A,C)**



$$3F = f \leq \mu_s Mg$$

$$3Kt^2 \leq \mu_s Mg$$

$$t^2 \leq \frac{\mu_s Mg}{3K}$$

$$t^2 = \frac{0.5 \times 150 \times 10}{3 \times 40}$$

$$t \leq 2.5 \text{ sec}$$

for $t > 2.5 \text{ sec}$

$$3Kt^2 - f_k = Ma$$

$$3Kt^2 - \mu_k Mg = Ma$$

$$\frac{3K}{M}t^2 - \mu_k g = a$$

$$t = 5 \text{ sec}$$

$$a = 16 \text{ m/s}^2$$

6. **Ans (A,B)**

$$x = 2 \sin 2t$$

$$v_x = 4 \cos 2t$$

$$a_x = -8 \sin 2t$$

$$y = t^3 - 4t$$

$$v_y = 3t^2 - 4$$

$$a_y = 6t$$

$$z = 5$$

$$v_z = a_z = 0$$

SECTION-II

1. **Ans (32.00)**

$$w = mgh$$

$$w = 320 \text{ J}$$

$$P = \frac{w}{t}$$

$$P = 32 \text{ watt}$$

2. **Ans (5.87)**

by conservation of energy

$$0 + Mgx = \frac{1}{2}Mv'^2 + \frac{1}{2}mv^2 + mg(4) + mg(x - 3)$$

$$v \cos 37 = v'$$

$$\therefore v = 5.87 \text{ m/s}$$

3. **Ans (100.00)**

Given $D = 1 \text{ m}$,

$$\omega_0 = 600 \text{ rev/min} = \frac{600 \times 2\pi}{60} = 20\pi \text{ rad/sec}$$

$$\omega_f = 0 \text{ in } 20 \text{ sec}$$

$$\text{therefor, } \omega_f = \omega_0 = -\alpha t$$

$$\Rightarrow \alpha = \frac{\omega_0}{t} = \frac{20\pi}{20} = \pi \text{ rad/sec}^2$$

so the no. of revolution before coming to rest is

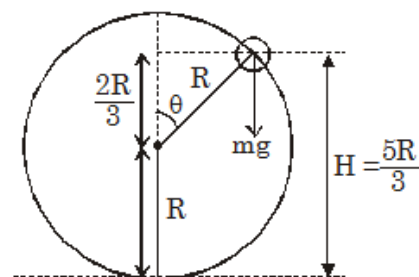
$$\omega_f^2 = \omega_0^2 - 2\alpha\theta$$

$$\theta = \frac{\omega_0^2}{2\alpha} = \frac{400\pi^2}{2 \times \pi} = 200\pi \text{ rad} = 100 \text{ rev}$$

4. **Ans (9.80)**

Block falls freely after leaving contact.

5. **Ans (0.00)**



$$R \cos \theta = \frac{2R}{3}$$

$$\cos \theta = \frac{2}{3}$$

$$\text{at } \theta = \cos^{-1} \left(\frac{2}{3} \right)$$

$$mg \cos \theta = \frac{mv^2}{R}$$

$$\text{So } N = 0$$

6. **Ans (16.00)**

$$\text{Energy} = (m)^x (a)^y (f)^z$$

$$ML^2T^{-2} = M^x (LT^{-2})^y (T^{-1})^z$$

$$ML^2T^{-2} = M^x L^y T^{-2y-z}$$

$$\Rightarrow x = 1, y = 2, z = -2$$

$$\Rightarrow \text{Energy} = \frac{ma^2}{f^2}$$

$$n_1 u_1 = n_2 u_2$$

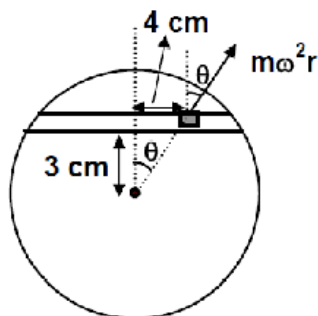
$$\frac{1 \text{ kg} (m/s^2)^2}{(\text{sec}^{-1})^2} = \frac{n_2 (10 \text{ gm}) (5 \text{ m/s}^2)^2}{(2 \text{ sec}^{-1})^2}$$

$$\Rightarrow n_2 = 16$$

SECTION-III

1. **Ans (2)**

Relative motion between block and table will start when



$$m\omega^2 r \sin \theta = \mu(mg + m\omega^2 r \cos \theta) \dots (i)$$

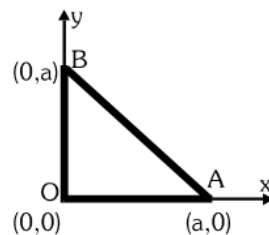
$$\omega = \alpha t \dots (ii)$$

$$\text{solving (i) and (ii) } t = \sqrt{200} \text{ sec.}$$

2. **Ans (2)**

$$\omega = \sqrt{\frac{\mu g}{R}} = 2 \text{ rad/sec}$$

3. **Ans (8)**



Centre of Mass of rod OA is at $(\frac{a}{2}, 0)$

Centre of Mass of rod OB is $(0, \frac{a}{2})$

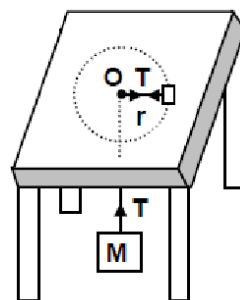
Centre of Mass of rod AB is $(\frac{a}{2}, \frac{a}{2})$

For system

$$X_{cm} = \frac{m \times \frac{a}{2} + m \times 0 + \sqrt{2}m \times \frac{a}{2}}{m + m + \sqrt{2}m} = \frac{a}{\sqrt{8}}$$

4. **Ans (6)**

For circular motion of a body tied to a string on a horizontal plane $\frac{mv^2}{r} = T$



here, $T = mg$

$$\frac{mv_1^2/r_1}{mv_2^2/r_2} = \frac{M_1 g}{M_2 g}$$

$$\frac{v_1^2 r_2}{v_2^2 r_1} = \frac{8}{1} \dots (i)$$

Now as angular momentum is conserved about pt O.

$$mv_1 r_1 = mv_2 r_2$$

$$\text{from (i) and (ii) } \frac{r_1}{r_2} = 2$$

$$\text{further } \frac{\omega_2}{\omega_1} = \frac{v_2}{v_1} \times \frac{r_1}{r_2} = \left(\frac{r_1}{r_2}\right)^2$$

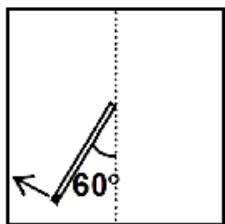
$$\frac{\omega_2}{\omega_1} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\frac{\Delta \omega}{\omega} = \frac{\omega_2 - \omega_1}{\omega_2} = \frac{\omega_2}{\omega_1} - 1 = \frac{1}{4} - 1 = -\frac{3}{4}$$

5. **Ans (8)**

$$(20 - \ell_0)k - (16 - \ell_0)k = m \times 400 \times 20$$

6. Ans (8)



Conservation of energy

$$\frac{1}{2}mv^2 - mg\frac{l}{2} = mg\frac{l}{2} + m\sqrt{3}\left(2l\frac{\sqrt{3}}{2}\right)g$$

$$v = \sqrt{8gl}$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A,D)

- (A) 1gm N_2O & 1 gm CO_2
(N = 14, C = 12, O = 16)
Molar mass of N_2O = 44 g/mol
Molar mass of CO_2 = 44 g/mole

$$\therefore \text{Moles} = \frac{\text{Mass}}{\text{Molar Mass}}$$

- $\therefore$ 1 molecule of N_2O contains $\Rightarrow$ 3 atoms.
 $\therefore$ 1 molecule of CO_2 contains $\Rightarrow$ 3 atoms.
{Both contain same no. of molecule or moles. Hence they contain same no. of atoms.}

Ans. (A)

- (B) 1 lit. 20%

$$\frac{W}{V} \left\{ \frac{W}{V} = \frac{\text{Weight of solute (gm)}}{\text{Volume of solution (mL)}} \times 100 \right\}$$

$$\therefore 20 = \frac{\text{Wt. of solute}}{1000} \times 100$$

$$\therefore \text{Wt. of solute} = 200g$$

$\Rightarrow$ 5m $\Rightarrow$ 5 moles of urea present in 1000 g of solvent

$\Rightarrow$ In 1000 g solvent

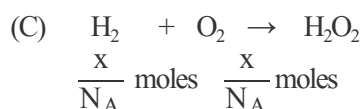
$\Rightarrow$ Mass of urea = (5 $\times$ Molar mass)

[Urea NH_2CONH_2]

$$= 5 \times 60 = 300 \text{ gm}$$

$$\therefore \text{Total mass of solution} = (1000 + 300) \\ = 1300 \text{ g} = 1.3 \text{ kg.}$$

$$\text{Mass of solute (urea)} = 300 \text{ gm}$$



$\Rightarrow$ Lets assume : x is the no of molecules of H_2 & O_2 .

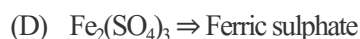
According to stoichiometry coefficient.

(molar ratio of H_2 & O_2 1 : 1) $\leftarrow$ (reacting)

$$\text{Given molar ratio} \Rightarrow \frac{x}{N_A} : \frac{x}{N_A} \Rightarrow 1 : 1$$

$\therefore$ Both are consumed after completion of reaction.

Ans. (C)



$$\therefore \%Fe = \frac{2 \times 56}{\text{M wt. of comp.}} \times 100$$

$$\text{M. wt. of comp.} = (56 \times 2) + (32 + 64) \times 3 = 400 \text{ g/mol}$$

$$\therefore \%Fe = \frac{2 \times 56}{400} \times 100 = 28\%.$$

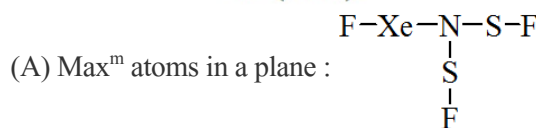
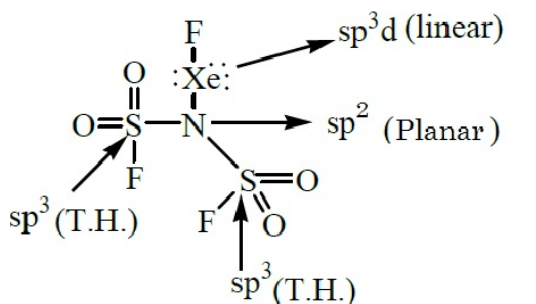
Ans. (A & D)

2. Ans (B,C)

$$x \times 9 = 0.4 \times 18 \times 0.5$$

$$x = 0.4$$

4. Ans (A,D)



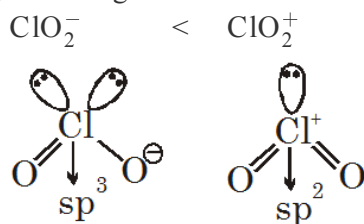
(B) central atom are sp^3 hybridized in structure (I)

(C) 5, d-orbitals are involved in bonding in structure (I). sp^3d ; four d-orbitals in π -bonding

(D) size of O < N

5. Ans (A,D)

(A) Bond angle

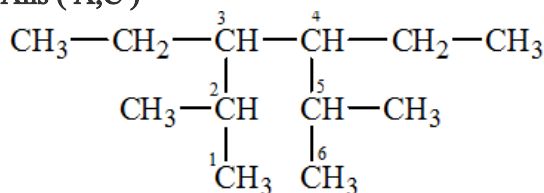


(B) dipole moment : $\text{:C}\equiv\text{O:} < \text{H}-\text{N}-\text{H}$

(C) Boiling point :- $\text{SbH}_3 > \text{NH}_3$

(D) Melting point : (VWF $\propto$ size) $\text{HI} > \text{HF} > \text{HBr} > \text{HCl}$
(due to zig-zag structure, there will be no H-bonds present between layers.)

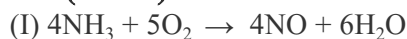
6. Ans (A,C)



Correct IUPAC nomenclature is 3, 4-diethyl 2, 5-dimethyl hexane

SECTION-II

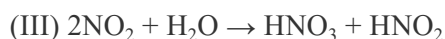
1. Ans (27.24)



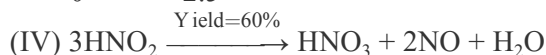
$$\begin{array}{cccc} 2 & 3 & & \\ 0 & 0.5 & 2 & 3 \end{array}$$



$$\begin{array}{ccc} 2 & 0.5 & 1 \\ 1 & 0 & \end{array}$$



$$\begin{array}{cccc} 1 & 3 & 0.5 & 0.5 \\ 0 & 2.5 & & \end{array}$$



$$\begin{array}{ccc} 0.5 & & \\ & \downarrow & \\ & \frac{2}{3} \times 0.5 \times 0.6 & \\ & = 0.2 \text{ mole} & \end{array}$$

$$\therefore \text{NO} = 1 + 0.2 = 1.2 \text{ mole} = 1.2 \times 22.7 = 27.24 \text{ L}$$

4. Ans (1.33)

(A) sp^2, sp^2

(B) sp^3

(C) sp^2

(D) sp^3

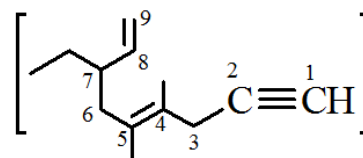
(E) sp

Type - I $\Rightarrow$ (x) $\text{sp}^3 \rightarrow (2)$

Type - II $\Rightarrow$ (y) $\text{sp}^2 \rightarrow (3)$

Type - III $\Rightarrow$ (z) $\text{sp} \rightarrow (1)$

5. Ans (93.16)



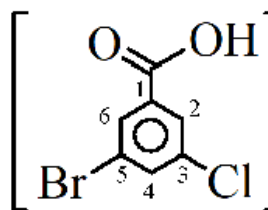
(x) = 9

(y) = 3

(z) = 16

Here 4 sp^2 Hybridised c-atoms and 2 sp hybridised c-atoms are present and each sp^2 and sp hybridised carbon contains 3 and 2 hybridised orbitals respectively.

6. Ans (26.18)



(x) = 2

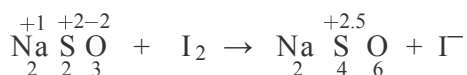
(y) = 6

(z) = 18

Here each sp^2 hybridised carbon atom contains 3 hybridised orbitals.

SECTION-III

1. **Ans (9)**



$$(n_f = 1) \quad (n_f = 2)$$

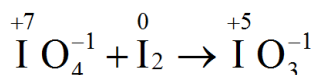
$$\therefore \text{Meq. of Na}_2\text{S}_2\text{O}_3 = \text{meq. of I}_2$$

$$\Rightarrow M \times V_{\text{mL}} \times n_f = \text{mmoles} \times n_f$$

$$\therefore \text{m moles of I}_2 = \frac{0.1 \times 100}{2} = 5$$

$$\therefore \text{m moles of I}_2 = 5 \rightarrow \text{from 1}^{\text{st}} \text{ reaction}$$

Similarly for 2nd reaction



$$n_f = 2 \quad n_f = 10$$

$$\Rightarrow \text{meq. of IO}_4^{-1} = \text{meq. of I}_2$$

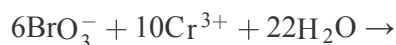
$$M \times V_{\text{mL}} \times n_f = (\text{m moles}) \times n_f$$

$$\therefore \text{m moles of I}_2 = \frac{0.2 \times 100 \times 2}{10}$$

$$\therefore \text{m moles of I}_2 = 4 \text{ from 2}^{\text{nd}} \text{ reaction}$$

$$\text{total m moles of I}_2 = 5 + 4 = 9 \text{ m moles}$$

2. **Ans (6)**



$$x = 6, y = 10, z = 22$$

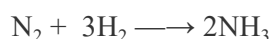
$$\Rightarrow (z - x - y) = 6$$

3. **Ans (1)**

$$[\text{Na}^+] + [\text{Ca}^{+2}] = \frac{(2 \times 100 \times 2) + (3 \times 100) + (1 \times 200)}{400}$$

$$[\text{SO}_4^{-2}] + [\text{Cl}^-] = \frac{2 \times 100 + 3 \times 100 + 1 \times 200 \times 2}{400}$$

4. **Ans (4)**



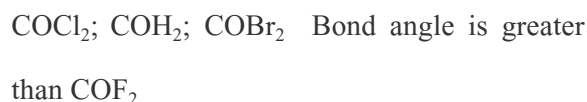
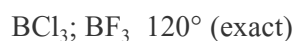
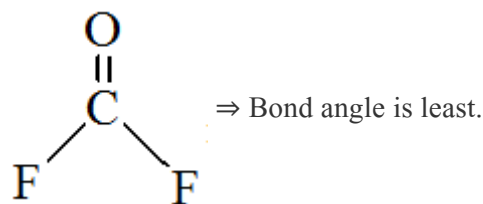
$$a \quad 3a$$

$$a \times 28 + 3a \times 2 = 340$$

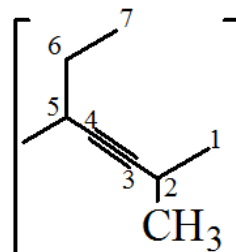
$$a = 10 \text{ mole}$$

$$\text{Mass of NH}_3 = 10 \times 2 \times \frac{20}{100} \times 17 = 68 \text{ gm}$$

5. **Ans (0)**



6. **Ans (7)**



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A,B,C,D)**

$$\cos(\alpha + \beta) + \sin(\alpha - \beta) = 0$$

$$\cos\alpha \cos\beta - \sin\alpha \sin\beta + \sin\alpha \cos\beta - \cos\alpha \sin\beta = 0$$

$$(\cos\beta - \sin\beta)(\cos\alpha + \sin\alpha) = 0$$

$$\text{Either } \cos\beta - \sin\beta = 0 \text{ or } \cos\alpha + \sin\alpha = 0$$

$$\text{If } \tan\beta = 1$$

$$\text{If } \tan\alpha = -1$$

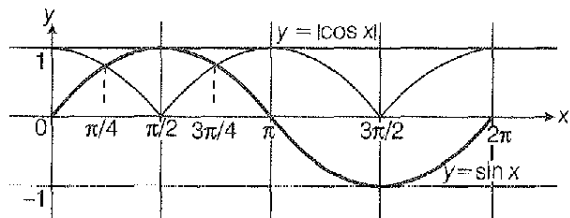
$$\text{then } \alpha \in \mathbb{R}$$

$$\text{then } \beta \in \mathbb{R}$$

$$\text{So } \alpha + \beta = \pi, \frac{3\pi}{2}, 5\pi, \frac{5\pi}{2}$$

2. **Ans (A,C)**

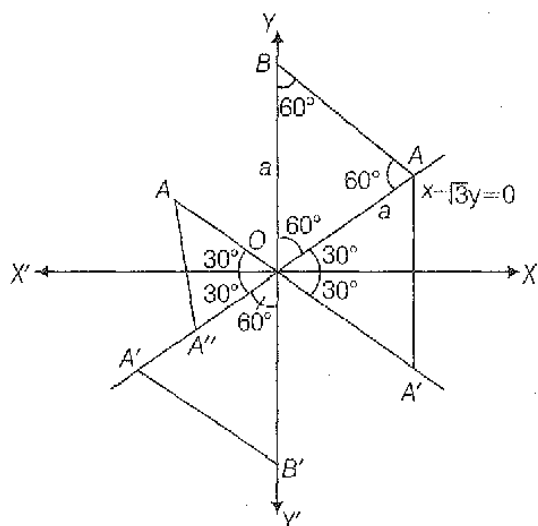
It is easier to solve the inequality using graphical method. The graphs of $y = |\cos x|$ and $y = \sin x$ are shown in the following figure.



From the figure, $|\cos x| \leq \sin x$ for $x \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

3. **Ans (A,B,C,D)**

If B lies on Y-axis, then coordinates of B are $(0, a)$ or $(0, -a)$

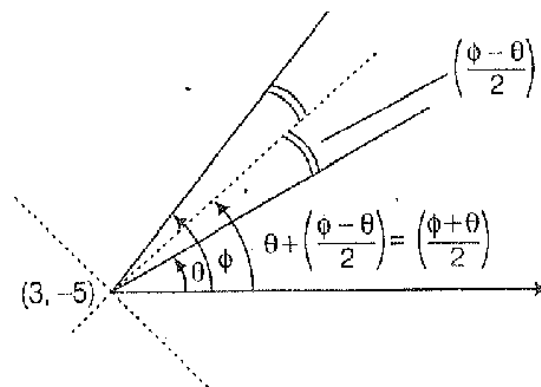


If third vertex in IV quadrant or in II quadrant, then its coordinates are $(a \cos 30^\circ, -a \sin 30^\circ)$ and $(-a \cos 30^\circ, a \sin 30^\circ)$

i.e., $\left(\frac{a\sqrt{3}}{2}, -\frac{a}{2}\right)$ and $\left(-\frac{a\sqrt{3}}{2}, \frac{a}{2}\right)$

4. **Ans (A,B,C,D)**

One bisector makes an angle $\left(\frac{\theta + \phi}{2}\right)$ with X-axis, then other bisector makes an angle $90^\circ + \left(\frac{\theta + \phi}{2}\right)$ with X-axis.



Equation of bisectors are

$$\frac{x-3}{\cos\left(\frac{\theta+\phi}{2}\right)} = \frac{y+5}{\sin\left(\frac{\theta+\phi}{2}\right)} \quad \dots(i)$$

$$\text{and } \frac{x-3}{\cos\left(\frac{\pi}{2} + \frac{\theta+\phi}{2}\right)} = \frac{y+5}{\sin\left(\frac{\pi}{2} + \frac{\theta+\phi}{2}\right)}$$

$$\Rightarrow \frac{x-3}{-\sin\left(\frac{\theta+\phi}{2}\right)} = \frac{y+5}{\cos\left(\frac{\theta+\phi}{2}\right)} \quad \dots(ii)$$

$$\text{But given bisector are } \frac{x-3}{\cos \alpha} = \frac{y+5}{\sin \alpha}$$

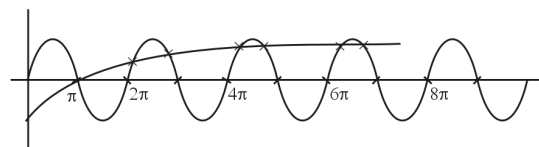
$$\therefore \alpha = \frac{\theta+\phi}{2} \text{ and } \frac{x-3}{\beta} = \frac{y+5}{\gamma} \quad [\text{from Eq. (i)}] \quad \dots(iii)$$

$$\therefore \beta = -\sin\left(\frac{\theta+\phi}{2}\right) = -\sin \alpha \quad [\text{from eq. (ii)}]$$

$$\text{And } \gamma = \cos\left(\frac{\theta+\phi}{2}\right) = \cos \alpha$$

5. **Ans (A,C)**

$$2\sin x = \log_5 x$$



$$\therefore n = 7$$

6. **Ans (A,C,D)**

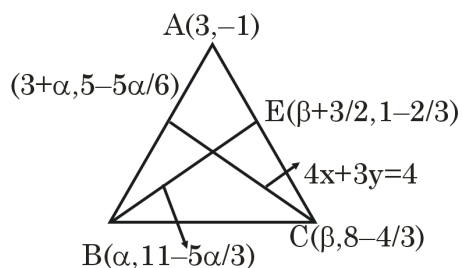
$A, B, C = B - \theta, B, B + \theta$, where θ is common difference.

$$\tan A \tan B \tan C = \tan(B - \theta) \tan B \tan(B + \theta)$$

$$= \frac{1 - \tan \theta}{1 + \tan \theta} \cdot \tan \frac{\pi}{4} \cdot \frac{1 + \tan \theta}{1 - \tan \theta} = 1$$

SECTION-II

1. Ans (5.00)



Point F lies on median CF

$$\therefore 4 \left(\frac{3+\alpha}{2} \right) + 3 \left(\frac{5-5\alpha}{6} \right) = 8$$

$\Rightarrow \alpha = 1$ and point E lies on median BE

$$\therefore 5 \left(\frac{B+3}{2} \right) + 3 \left(\frac{1-2B}{3} \right) = 11$$

$$\Rightarrow \beta = 5$$

$$\therefore B = (1, 2) \text{ and } C = (5, -4)$$

$$\therefore \text{equation of BC is } 3x + 2y = 7$$

2. Ans (2.00)

$$(13\ell + 19m - 32) + n(11\ell + 17m - 28) = 0$$

$$\text{Let } (13x + 19y - 32) + n(11x + 17y - 28) = 0$$

Equation is family of line passes through (1, 1)

$$p + q = 2$$

3. Ans (4.33)

$$\text{Let } A \equiv (3, 4), B \equiv (0, y), C \equiv (x, 0), D \equiv (8, 2)$$

$$\therefore \text{Slope of AB} = -\text{slope of BC}$$

$$\Rightarrow \frac{y-4}{0-3} = -\left(\frac{0-y}{x-0}\right)$$

$$\text{or } 4x - xy = 3y \quad \dots(i)$$

$$\text{and slope of BC} = -\text{slope of CD}$$

$$\Rightarrow \frac{y-4}{0-3} = -\left(\frac{0-y}{x-0}\right)$$

$$\text{or } 2x + xy = 8y \quad \dots(ii)$$

Adding equation (i) and (ii), we get

$$6x = 11y \quad \dots(iii)$$

From equation (ii) and (iii), we get

$$x = \frac{13}{3} = 4\frac{1}{3}$$

4. Ans (5.00)

Here, B is the image of A w.r.t. line $y = x$

$B \equiv (2, 1)$ and C is the image of A w.r.t.

line $x - 2y + 1 = 0$

$$\text{If } C \equiv (\alpha, \beta), \text{ then } \frac{\alpha-1}{1} = \frac{\beta-2}{-2} = \frac{-2(1-4+1)}{1+4}$$

$$\text{or } \alpha = \frac{9}{5} \text{ and } \beta = \frac{2}{5}$$

$$\therefore C \equiv \left(\frac{9}{5}, \frac{2}{5} \right)$$

$$\Rightarrow \text{Equation of BC is } y - 1 = \frac{\left(\frac{2}{5} - 1\right)}{\left(\frac{9}{5} - 2\right)}(x - 2)$$

$$\text{or } 3x - y - 5 = 0 \quad (\because \text{Eq. of BC is } ax + by - 5 = 0)$$

$$\text{Here } a = 3, b = -1 \quad \therefore a - 2b = 5$$

5. Ans (0.56)

We have, $a + b + c = 11$, $ab + bc + ca = 38$ and

$$abc = 40$$

$$\begin{aligned} \therefore \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} &= \frac{\left(\frac{b^2+c^2-a^2}{2bc}\right)}{a} + \frac{\left(\frac{c^2+a^2-b^2}{2ca}\right)}{b} + \frac{\left(\frac{a^2+b^2-c^2}{2ab}\right)}{c} \\ &= \frac{b^2+b^2-a^2}{2abc} + \frac{c^2+a^2-b^2}{2abc} + \frac{a^2+b^2-c^2}{2abc} \\ &= \frac{a^2+b^2-c^2}{2abc} = \frac{(a+b+c)^2 - 2(ab+bc+ca)}{2abc} \\ &= \frac{(11)^2 - 2(38)}{2 \times 40} = \frac{121 - 76}{80} = \frac{45}{80} = \frac{9}{16} = 0.56 \end{aligned}$$

6. Ans (8.50)

$$\begin{aligned} \sum_{r=1}^{18} \cos^2(5r)^\circ &= \cos^2 5^\circ + \cos^2 10^\circ + \cos^2 15^\circ + \dots + \cos^2 85^\circ + \cos^2 90^\circ \\ &= (\cos^2 5^\circ + \cos^2 85^\circ) + (\cos^2 10^\circ + \cos^2 80^\circ) + (\cos^2 15^\circ + \cos^2 75^\circ) \\ &\quad + \dots + (\cos^2 40^\circ + \cos^2 50^\circ) + \cos^2 45^\circ \\ &= (\cos^2 5^\circ + \sin^2 5^\circ) + (\cos^2 10^\circ + \sin^2 10^\circ) + (\cos^2 15^\circ + \sin^2 15^\circ) \\ &\quad + \dots + (\cos^2 40^\circ + \sin^2 40^\circ) + \cos^2 45^\circ \\ &= 1 + 1 + 1 + \dots + 1 + \frac{1}{2} = 8 + \frac{1}{2} = \frac{17}{2} \end{aligned}$$

SECTION-III

1. Ans (3)

The equation of straight line through (2, 3) with slope m is $y - 3 = m(x - 2)$ or $mx - y = 2m - 3$

$$\text{or } \frac{x}{\left(\frac{2m-3}{m}\right)} + \frac{y}{(3-2m)} = 1$$

$$\text{Here, } OA = \frac{2m-3}{m} \text{ or } OB = 3-2m$$

The area of $\Delta OAB = 12$

$$\Rightarrow \left| \frac{1}{2} \times OA \times OB \right| = 12$$

$$\text{or } \frac{1}{2} \left(\frac{2m-3}{m} \right) (3-2m) = \pm 12$$

$$\text{or } (2m-3)^2 = \pm 24m$$

Taking positive sign,

$$\text{we get } 4m^2 - 36m + 9 = 0$$

Here $D > 0$, This is a quadratic in m which given two value of m, and taking negative sign,

$$\text{we get } (2m+3)^2 = 0.$$

$$\text{This gives one line of m as } \frac{-3}{2}.$$

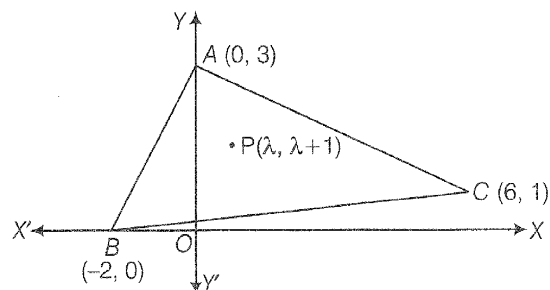
Hence, three straight lines are possible.

2. Ans (2)

$$\text{Equation of AB is } 3x - 2y + 6 = 0$$

$$\text{Equation of BC is } 8 - 8y + 2 = 0,$$

$$\text{Equation of CA is } x + 3y - 9 = 0$$



$$\text{Let } P \equiv (\lambda, \lambda + 1)$$

$\because$ B and P lie on one side of AC, then

$$\frac{\lambda + 3(\lambda + 1) - 9}{-2 + 0 - 9} > 0 \text{ or } 4\lambda - 6 < 0$$

$$\text{or } \lambda < \frac{3}{2} \quad \dots(i)$$

and C and P lie on one side of AB, then

$$\frac{3\lambda - 2(\lambda + 1) + 6}{18 - 2 + 6} > 0 \text{ or } \lambda + 4 > 0$$

$$\text{or } \lambda > -4 \quad \dots(ii)$$

Finally, A and P lie on one side of BC, then

$$\frac{\lambda - 8(\lambda + 1) + 2}{0 - 24 + 2} > 0 \text{ or } -7\lambda - 6 < 0$$

$$\text{or } \lambda > -\frac{6}{7} \quad \dots(iii)$$

From Eqs. (i), (ii) and (iii),

$$\text{we get } -\frac{6}{7} < \lambda < \frac{3}{2}$$

Integral value of λ are 0 and 1.

Hence, number of integral value of λ is 2.

3. Ans (4)

$$\underbrace{(4\cos^2 x + \sec^2 x)}_{\geq 4} + \underbrace{(\tan^2 x + \cot^2 x)}_{\geq 2} = 6$$

$$\therefore \cos^2 x = \frac{1}{2} \text{ and } \tan^2 x = 1$$

$$\therefore \cos x = \pm \frac{1}{2}$$

Hence 4 solution

4. **Ans (1)**

$$\sec 610^\circ \operatorname{cosec} 160^\circ - \cot 380^\circ \tan 470^\circ$$

$$= -\sec 70^\circ \operatorname{cosec} 20^\circ + \cot 20^\circ \tan 70^\circ$$

$$= -\frac{1}{\sin 20^\circ \cos 70^\circ} + \frac{\sin 70^\circ \cos 20^\circ}{\sin 20^\circ \cos 70^\circ}$$

$$= -\frac{(1 - \sin^2 70^\circ)}{\sin^2 20^\circ} = -\frac{\cos^2 70^\circ}{\sin^2 20^\circ} = -1$$

5. **Ans (7)**

$$\left. \begin{array}{l} a - 4b + c = 0 \\ \& ap + bq + c = 0 \end{array} \right\} \Rightarrow p = 1, q = -4$$

6. **Ans (2)**

As we know that

$$\frac{1 - \sin x}{\cos x} = \frac{\cos x}{1 + \sin x}$$

$$\log \left| \frac{1 - \sin x}{\cos x} \right| = \log \left| \frac{1 + \sin x}{\cos x} \right| = -1$$

Now, series is

$$\text{Let } S = 1 - \frac{x}{2} - \frac{x^2}{4} - \dots = 1 - \frac{\frac{x}{2}}{1 - \frac{x}{2}}$$

$$= 1 - \frac{-x}{2-x} = \frac{2-x-x}{2-x} = \frac{2(1-x)}{(2-x)} = \frac{k(1-x)}{(2-x)}$$

Thus, $k = 2$

JEE(Main + Advanced) : NURTURE COURSE
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	A	B	B	D	B	A	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	B	D	C	B	A	B	C	A
	Q.	1	2	3	4	5	6	7	8	9	10
A.	6.00	1.28	0.00	5.00	2.00	3.00 to 3.10	20.00	90.00	400.00	2.18 to 2.20	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	B	A	D	C	D	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	D	B	B	A	A	A	D	A	B
	Q.	1	2	3	4	5	6	7	8	9	10
A.	1.60	1.00	1.00	4.00	68.10	8.00	8.00	1.00	3.00	3.00	

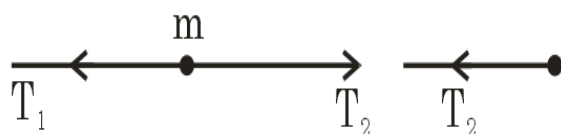
PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	B	A	C	A	B	A	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	A	D	C	B	D	C	D	A	A
	Q.	1	2	3	4	5	6	7	8	9	10
A.	23.00	2.00	11.00	2.00	-4.00	40.00	5.00	1.00	-19.00	468.75	

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (B)

$$T_1 - T_2 = m\omega^2 r_1$$

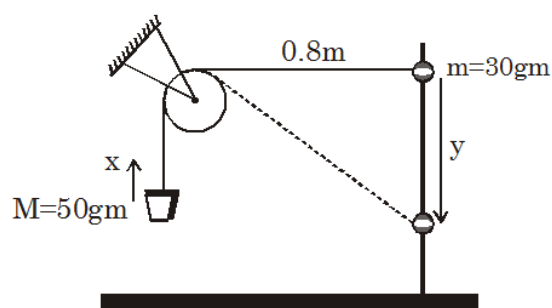


$$T_2 = m\omega^2(r_1 + r_2)$$

$$T_1 = m\omega^2(2r_1 + r_2)$$

 Thus, $T_1 > T_2$

2. Ans (A)



$$30 gy = 50 gx \Rightarrow 3y = 5x \dots (1)$$

$$\sqrt{0.64 + y^2} - 0.8 = x$$

$$y^2 = x^2 + 1.6x$$

$$16x^2 = 1.6x \times 9$$

$$x = 0.9 \text{ m} \Rightarrow y = 1.5 \text{ m}$$

3. **Ans (A)**

$$\frac{1}{2}mv^2 = bd^2$$

$$ma_r = \frac{mv^2}{1} = 2bd^2 = 6b$$

$$\frac{m}{2}2v \frac{dv}{dt} = b2dv$$

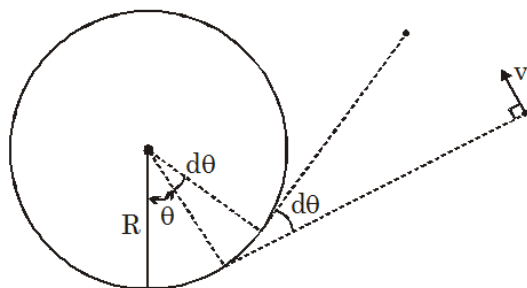
$$ma_t = 2bd = 2\sqrt{3}b$$

$$F = ma_T = \sqrt{48}b = 4\sqrt{3}b$$

4. **Ans (C)**

$$mg \times \cos 60^\circ = \int_0^x kx dx$$

5. **Ans (B)**



$$\omega = \frac{v}{\ell - R\theta}$$

$$\int_0^{\theta_0} (\ell - R\theta) d\theta = \int_0^t v dt$$

$$\theta_0 = \frac{4}{1} \left(\ell\theta - \frac{R\theta^2}{2} \right) = vt$$

$$16 - 8 = 2t \Rightarrow t = 4 \text{ sec}$$

6. **Ans (D)**

$$\frac{1}{2} \times 16 = 0.4 \times 10 \times (0.25 + x) + \frac{1}{6}6x^2$$

$$8 = 1 + 4x + 3x^2$$

$$3x^2 + 4x - 7 = 0$$

$$x = 1$$

7. **Ans (B)**

First law of motion.

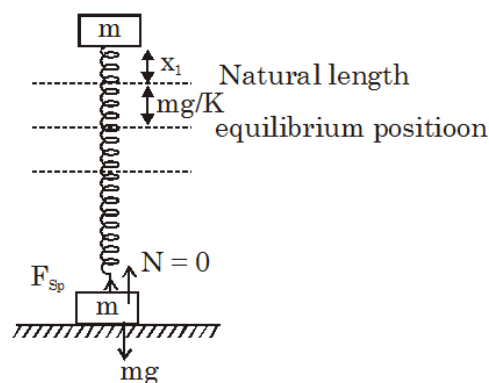
8. **Ans (A)**

$$W_{fr} + W_{mg} = 0 \Rightarrow W_{fr} = -mgh$$

9. **Ans (C)**

Non uniform Circular motion

10. **Ans (C)**



$$F_{Sp} = Kx_1 = mg \Rightarrow x_1 = \frac{mg}{K}$$

$$\text{Compression} = \frac{3mg}{K}$$

11. **Ans (B)**

$$W = \Delta K$$

$$50 \times 2 - \frac{1}{2}5(25 - 1) = \frac{1}{2}20V^2$$

$$100 - 60 = 10 V^2$$

$$V^2 = 4 \Rightarrow V = 2 \text{ m/s}$$

12. **Ans (B)**

$$mg \left(\frac{L}{2} + y - L \cos \theta \right) = \frac{1}{2}m 5gy$$

$$\frac{L}{2} - L \cos \theta = \frac{3y}{2} \dots (i)$$

$$\text{Also } \frac{L}{\sqrt{2}} + y = L \Rightarrow y = L - \frac{L}{\sqrt{2}} \dots (ii)$$

$$\text{from (i) } L - 2L \cos \theta = 3L - \frac{3L}{\sqrt{2}}$$

$$\cos \theta = \frac{\frac{3}{\sqrt{2}} - 2}{2} = 0.06$$

13. **Ans (B)**

$$mgR = \frac{1}{2}mv_1^2 + \frac{1}{2}m \frac{16v_1^2}{9}$$

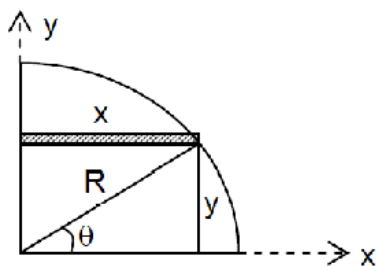
$$gR = \frac{25v_1^2}{18}$$

$$v_1 = \frac{6}{\sqrt{5}} \approx 2.68 \text{ m/s}$$

14. **Ans (D)**

Force is always perpendicular to velocity

15. Ans (C)



elementary mass = $x \, dy$

$$x = R \cos \theta, y = R \sin \theta$$

$$dy = R \cos \theta d\theta$$

$$\text{elementary mass} = \sigma(R \cos \theta)(R \cos \theta d\theta)$$

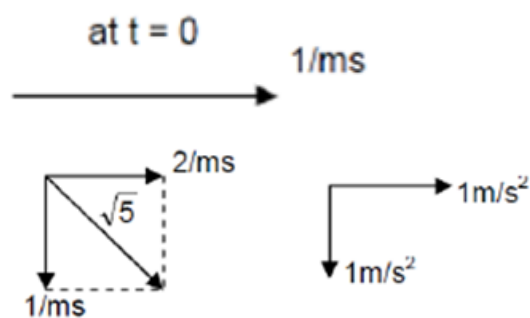
$$\sigma R^2 \cos^2 \theta d\theta$$

$$y_{\text{an}} = \frac{\int y \, dm}{\int \cdot dm} = \frac{\int_0^{\frac{\pi}{2}} R \cos \theta \sigma R^2 \cos^2 \theta d\theta}{\int_0^{\frac{\pi}{2}} \sigma R^2 \cos^2 \theta d\theta}$$

$$= \frac{4}{3\pi} R$$

16. Ans (B)

at $t = 1 \text{ sec}$



$$R = \frac{V^2}{a_n} = \frac{(\sqrt{5})^2}{\frac{1}{\sqrt{5}}} = 5\sqrt{5} \text{ m}$$

17. Ans (A)

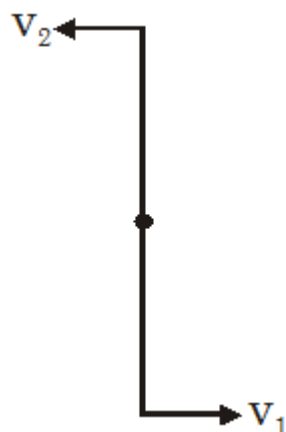
$$1.2gL \frac{3}{4} - 3g \times \frac{3}{5} \times \frac{L}{4}$$

$$= \frac{1}{2} 3v_1^2 \frac{9}{25} + \frac{1}{2} 1.2v_1^2$$

$$\frac{9L}{2} = 1.14v_1^2$$

$$v_1 = \frac{3}{2\sqrt{1.14}} = 1.4$$

18. Ans (B)



$$v_2^2 + 4g\ell = v_1^2$$

$$\frac{v_1}{v_2} = 2$$

$$v_2^2 + ug\ell = 4v_2^2$$

$$v_2 = \sqrt{\frac{4g\ell}{3}}$$

19. Ans (C)

$$h_{\text{max}} = \frac{3L}{2}$$

20. Ans (A)

$$-\mu mgt_0 = \frac{mv_0}{2} - mv_0$$

$$\mu = \frac{v_0}{2gt_0}$$

SECTION-II

1. Ans (6.00)

$$\frac{m_1 \frac{h}{4} - m_2 \frac{h}{4}}{m_1 - m_2} = h' \Rightarrow h' = \frac{h}{3}$$

2. Ans (1.28)

$$\tan \theta_0 = \frac{\omega^2 \sin \theta_0 r}{g}$$

$$\sec \theta_0 = \frac{(2 \times 2\pi)^2}{10} \times \frac{8}{100}$$

$$= \frac{128\pi^2}{1000} = 1.28$$

3. Ans (0.00)

no external force is present so center of mass will be at rest

4. **Ans (5.00)**

$$T + T \cos \theta = m\omega^2 r$$

$$T \sin \theta = mg$$

$$\frac{1 + \cos \theta}{\sin \theta} = \frac{\omega^2 r}{g} \quad \dots (i)$$

$$\sqrt{r^2 + 1} + r = \sqrt{5}$$

$$r^2 + 1 = 5 + r^2 - 2\sqrt{5}r$$

$$r = \frac{2}{\sqrt{5}}$$

$$\tan \theta = \frac{\sqrt{5}}{2}$$

Using (i)

$$\frac{1 + \frac{2}{3}}{\frac{\sqrt{5}}{3}} = \frac{\omega^2 2}{\sqrt{5} \times 10}$$

$$\omega = 5$$

5. **Ans (2.00)**

$$f = 2N$$

$$a_{\text{lk}} = 2 \text{ m/s}^2$$

$$S = \frac{1}{2} \times 2 \times 1 = 1 \text{ m}$$

$$W = 2J$$

6. **Ans (3.00 to 3.10)**

$$45 \omega^2 0.45 = 15 \omega^2 0.3 + 60 \times 0.25 \times 10$$

$$\omega^2 45 \times 35 = 6 \times 2500$$

$$\omega = \frac{10\sqrt{2}}{\sqrt{21}} \approx 3.1 \text{ rad/sec}$$

7. **Ans (20.00)**

Work done by friction

$$= \int \vec{F} \cdot d\vec{s} = \int_0^x \mu mg \cos \theta \frac{dx}{\cos \theta} = \mu mgx = 20 \text{ J}$$

8. **Ans (90.00)**

$$F = 10 x^2$$

$$\omega = 10 \times \frac{27}{3} = 90$$

9. **Ans (400.00)**

$$\frac{\ell}{\sin 30^\circ} = \frac{\ell'}{\sin 60^\circ}$$

$$\ell' = \sqrt{3}\ell$$

$$\ell' \omega' \cos 30^\circ = v \cos 30^\circ$$

$$\ell \omega = \cos 60^\circ (v + \ell' \omega') = \cos 60^\circ (2v) = v$$

$$\omega = v/\ell = \frac{40\sqrt{3}}{0.1\sqrt{3}} = 400 \text{ rad/s}$$

10. **Ans (2.18 to 2.20)**

$$\frac{m}{\ell} \times 0.4g \times 0.6 = \frac{1}{2}mv^2$$

$$v = \sqrt{4.8}$$

$$v = 2.2$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (B)**

Given $M_Y = 12 M_X$ (M = atomic mass)

	Mass	Mole	Simplest ratio	Whole no.
Y	7.2	$\frac{7.2}{M_Y} = \frac{7.2}{12 M_X} = \frac{6}{M_X}$	1 = 1	1 × 3 = 3
X	1	$\frac{1}{M_X}$	$\frac{1}{6} = \frac{10}{6} = \frac{5}{3}$	$\frac{5}{3} \times 3 = 5$
E.F. = Y_3X_5				

2. **Ans (B)**

$$\alpha = \frac{M_T - M_0}{M_0(n-1)} \Rightarrow 0.1 = \frac{M_A - 120}{120 \times 1}$$

$$\Rightarrow 12 = M_A - 120 \Rightarrow M_A = 132$$

3. **Ans (B)**

Ans. (2)

5. **Ans (D)**

$$m = \frac{1000}{78} = 12.82 \text{ m}$$

6. **Ans (C)**

$$X_{\text{solvent}} = x, X_{\text{solute}} = 1 - x$$

$$\text{Molality} = \frac{(1-x)}{x \times 18} \times 1000$$

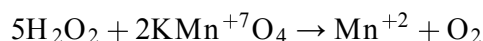
7. **Ans (D)**

$$n_{\text{CaCl}_2} = \frac{1.11 \times 10^{-3}}{111} = 10^{-5} = n_{\text{CaCO}_3}$$

$$W_{\text{CaCO}_3} = 10^{-5} \times 100 = 10^{-3} \text{ gm}$$

$$\text{Hardness} = \frac{10^{-3}}{1000} \times 10^{-6} = 1 \text{ ppm}$$

9. **Ans (B)**



$$n = 2 \quad n = 5$$

$$n_{\text{H}_2\text{O}_2} = \frac{5}{2}y$$

10. **Ans (B)**

(i) $\rightarrow$ (d) $(n+1)$ rule

(ii) $\rightarrow$ (c) two electron must have opposite spin

(iii) $\rightarrow$ (b) electron must be spin singly occupied

(iv) $\rightarrow$ (a, e) It was used by Bohr's and it represent size of orbit.

15. **Ans (A)**

Between H_2O and NH_3 , O-H bond being more polar, form H-bond with lone pair of NH_3 .

17. **Ans (A)**

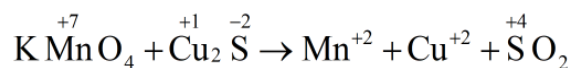
$$\mu : \text{BF}_3 = \text{BCl}_3 = \text{BBBr}_3 = \text{F}_2 = \text{Cl}_2 = \text{Br}_2 = 0$$

19. **Ans (A)**

Use $(n+1)$ rule

SECTION-II

1. **Ans (1.60)**

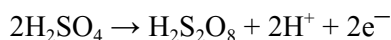


$$nf = 5 \quad nf = 2+6=8$$

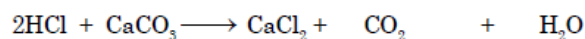
$$5 \times n = 8 \times 1$$

$$n = \frac{8}{5} = 1.6$$

3. **Ans (1.00)**



4. **Ans (4.00)**

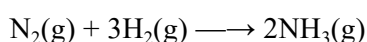


$$2.5 \text{ gm}$$

$$n = \frac{1.1}{44} = 0.025$$

$$\text{Mass} = \frac{2.5}{62.5} \times 100 = 4\%$$

5. **Ans (68.10)**



$$6 \quad 9$$

$$3 \quad - \quad 6 \times 1/2 = 3$$

$$\text{Volume} = 3 \times 22.7 = 68.1$$

6. **Ans (8.00)**

$$\% \frac{W}{W} = \frac{W_{\text{solute}}}{W_{\text{soln}}} \times 100$$

$$20 = \frac{x}{40} \times 100$$

$$x = 8 \text{ gm}$$

8. **Ans (1.00)**

sp^3 hybridised P atom = 3 sp^2 hybridised N atom = 3

$$\text{Ratio} = \frac{3}{3} = 1$$

9. **Ans (3.00)**

$d\pi - p\pi$ is present in : ClO_3^- , PO_4^{3-} , SO_3

10. **Ans (3.00)**

BF_3 , SF_6 , $\text{B}_3\text{N}_3\text{H}_6$, PCl_3F_2

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**

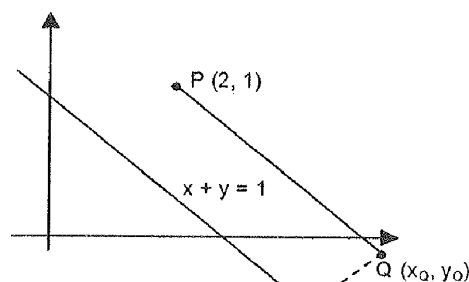
$$\text{slope of req. line} = \frac{-1}{\left(\frac{1}{-1}\right)} = 1$$

$$\therefore \text{line is } y - 1 = x - 3$$

$$\Rightarrow y = x - 2$$

2. **Ans (D)**

$$\text{PQ} = 3\sqrt{2}$$



$$\Rightarrow X_Q = 2 + 3\sqrt{2} \cdot \frac{1}{\sqrt{2}} = 5$$

$$\text{and } Y_Q = 1 - 3\sqrt{2} \cdot \frac{1}{\sqrt{2}} = -2$$

{use parametric coordinates}

$$\Rightarrow Q \equiv (5, -2)$$

3. **Ans (B)**

$$f(x, y) = 2x + 3y - 6$$

$$\Rightarrow f(\alpha, 2 + \alpha) = 5\alpha$$

$$f\left(\frac{3}{2}\alpha, \alpha^2\right) = 3\alpha + 3\alpha^2 - 6$$

$$\text{For the given condition } 5\alpha(3\alpha + 3\alpha^2 - 6) < 0$$

$$\text{or } \alpha(\alpha + 2)(\alpha - 1) < 0$$

$$\Rightarrow \alpha \in (-\infty, -2) \cup (0, 1)$$

4. **Ans (A)**

$$\left(\cos^2 x + \frac{1}{\cos^2 x}\right)(1 + \tan^2 2y)(3 + \sin 3z) = 4$$

$$\text{Since, } \cos^2 x + \frac{1}{\cos^2 x} \geq 2,$$

$$1 + \tan^2 y \geq 1, 2 \leq 3 + \sin 3z \leq 4$$

So, the only possibility is

$$\cos^2 x + \frac{1}{\cos^2 x} = 2, 1 + \tan^2 y = 1,$$

$$3 + \sin 3z = 2$$

$$\Rightarrow \cos x = \pm 1 \Rightarrow x = n\pi$$

$$\tan 2y = 0 \Rightarrow y = \frac{m\pi}{2}$$

$$\sin 3z = -1 \Rightarrow z = (4k - 1)\frac{\pi}{6}; m, n, k \in \mathbb{I}.$$

5. **Ans (C)**

Here, $x + 2y - 3 = 0$ and $3x + 4y - 7 = 0$ intersect

$(1, 1)$, which does not satisfy $2x + 3y - 4 = 0$ and

$4x + 5y - 6 = 0$. Also, $3x + 4y - 7 = 0$ and $2x + 3y$

$-4 = 0$ intersect at $(5, -2)$ which does not satisfy x

$+ 2y - 3 = 0$ and $4x + 5y - 6 = 0$

Intersection point of $x + 2y - 3 = 0$ and $2x + 3y - 4$

$= 0$ is $(-1, 2)$ which satisfy $4x + 5y - 6 = 0$.

Hence, only three lines are concurrent.

6. **Ans (A)**

$$\tan \alpha \tan 2\alpha \tan 3\alpha \dots \tan(2n - 1)\alpha$$

$$= \tan(\alpha) \cdot \tan(2\alpha) \dots \tan(2n - 2)\alpha \cdot \tan(2n - 1)\alpha$$

$$= \tan(\alpha) \cdot \tan(2\alpha) \dots \tan(n\alpha) \dots$$

$$\dots \tan(2n\alpha - 2\alpha) \cdot \tan(2n\alpha - \alpha)$$

$$\therefore 4n\alpha = \pi$$

$$\therefore 2n\alpha = \frac{\pi}{2} \text{ \& } n\pi = \frac{\pi}{4}$$

$$\Rightarrow \tan(\alpha) \tan(2\alpha) \dots \tan\left(\frac{\pi}{4}\right) \dots$$

$$\dots \tan\left(\frac{\pi}{2} - 2\alpha\right) \cdot \tan\left(\frac{\pi}{2} - \alpha\right)$$

$$= \tan(\alpha) \cdot \tan(2\alpha) \dots 1 \dots \cot(2\alpha) \cot(\alpha)$$

$$= 1$$

7. **Ans (B)**

$$\sin 10^\circ + \sin 50^\circ - \sin 70^\circ$$

$$= 2\sin 30^\circ \cos 20^\circ - \sin 70^\circ$$

$$= 0$$

8. **Ans (A)**

$$4\sin^2 x + 3\cos^2 x + \sin \frac{x}{2} + \cos \frac{x}{2}$$

$$= 3 + \sin^2 x + \sin \frac{x}{2} + \cos \frac{x}{2}$$

$$\text{Now } \sin \frac{x}{2} + \cos \frac{x}{2} \text{ is maximum at } \frac{x}{2} = \frac{\pi}{4}$$

$$\text{or } x = \frac{\pi}{2}$$

and its maximum value is $\sqrt{2}$

At $x = \frac{\pi}{2}$, $\sin^2 x$ is also maximum and its maximum value is 1

$$\therefore 3 + \sin^2 x + \sin \frac{x}{2} + \cos \frac{x}{2} \text{ has maximum}$$

$$\text{value} = 3 + 1 + \sqrt{2} = 4 + \sqrt{2}$$

9. **Ans (A)**

In any triangle ABC

$$\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$$

$$c + a + a^2 = 1$$

$$a + a^2 = 1 - c$$

10. **Ans (A)**

$$a(x + y + 1) + b(2x - 3y - 8) = 0 \text{ passes through}$$

the intersection of $x + y + 1 = 0$ and $2x - 3y - 8$

$= 0$, i.e., passes through the point $(1, -2)$ and

$$\lambda(x - y - 1) + \mu(2x - y - 1) = 0 \text{ passes through}$$

always $(0, -1)$

$\therefore$ common member of both the family passes

through $(1, -2)$ and $(0, -1)$, $x + y + 1 = 0$

11. **Ans (A)**

$$\begin{vmatrix} 2 & 3 & 1 \\ 3 & 1 & -2 \\ a & 2 & -b \end{vmatrix} = 0$$

$$\Rightarrow b - a + 2 = 0$$

$$\Rightarrow a - b - 2 = 0$$

Now the given equation $ax^2 - bx - 2 = 0$

$$\Rightarrow (b+2)x^2 - bx - 2 = 0$$

$$D = b^2 + 8(b+2)$$

$$= b^2 + 8b + 16 = (b+4)^2 > 0$$

So roots are real.

12. **Ans (A)**

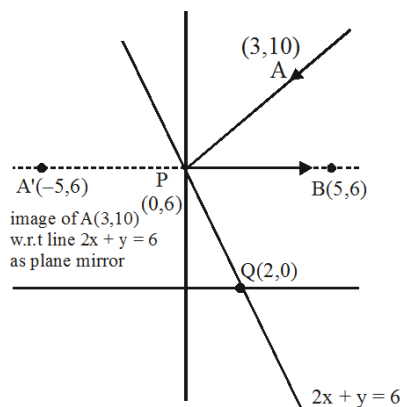
Mirror image of A(3, 10) if in line mirror

$2x + y - 6 = 0$ be (h, k), then

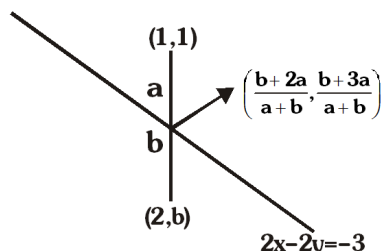
$$\frac{h-3}{2} = \frac{k-10}{1} = \frac{-2(6+10-6)}{4+1}$$

$$h = -5, k = 6$$

$\Rightarrow$ Equation of reflected beam is $y = 6$



13. **Ans (D)**



Point $\left(\frac{b+2a}{a+b}, \frac{b+3a}{a+b}\right)$ lie on $2x - 2y + 3 = 0$

$$2b + 4a - 2b - 6a + 3a + 3b = 0$$

$$\Rightarrow -2a + 3a + 3b = 0$$

$$\Rightarrow a + 3b = 0$$

$$\Rightarrow \frac{a}{b} = -3$$

14. **Ans (C)**

Using

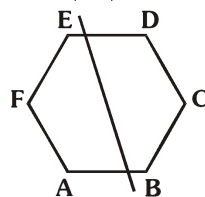
$$\frac{3 - \tan^2 \theta}{1 - 3 \tan^2 \theta} = \frac{\tan 3\theta}{\tan \theta}$$

$$\frac{\tan 3\theta}{\tan \theta} \times \frac{\tan 3^2 \theta}{\tan 3\theta} \times \dots \times \frac{\tan 3^n \theta}{\tan 3^{n-1} \theta} = \frac{\tan 3^n \theta}{\tan \theta}$$

$$\alpha = 3^n \theta \text{ \& } \beta = \theta$$

$$\frac{\alpha + \beta}{\theta} = (3^n + 1)$$

15. **Ans (B)**



Vertices B, C, D line on one side & A, E, F lies on other side of the given line hence divides into two pentagons.

16. **Ans (D)**

$$\begin{aligned} \frac{\cot C}{\cot A + \cot B} &= \frac{\cos C \sin A \sin B}{\sin C \sin(A+B)} \\ &= \frac{\cos C \sin A \sin B}{\sin^2 C} \\ &= \frac{a^2 + b^2 - c^2}{2ab} \times \frac{ab}{c^2} = \frac{a^2 b^2 - c^2}{2c^2} \\ &= \frac{670}{2c^2} = 335 \end{aligned}$$

17. **Ans (C)**

Required point is the foot of perpendicular from (1, 4)

$$\text{on } x - y + 1 = \frac{x-1}{1} = \frac{y-4}{-1} = \frac{(1-4+1)}{2}$$

$$x - 1 = -y + 4 = 1$$

$$x = 2, y = 3$$

18. **Ans (D)**

In equilateral triangle $r = \frac{R}{2} \Rightarrow R \in Q$

$$r_1 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = 4R \cdot \frac{1}{2} \left(\frac{\sqrt{3}}{2}\right)^2 \in Q$$

$$\Delta = \frac{abc}{4R} = 2R^2 \sin A \sin B \sin C = 2R^2 \left(\frac{\sqrt{3}}{2}\right)^3$$

$$\Rightarrow \Delta \notin Q$$

$$\text{Perimeter} = a + b + c = 2R (\sin A + \sin B + \sin C)$$

$$= 2R \cdot 3 \frac{\sqrt{3}}{2} = 3\sqrt{3}R$$

$$\text{Perimeter} \notin Q$$

19. **Ans (A)**

The equation of the bisectors of the angles between the lines are $\frac{x-2y+4}{\sqrt{1+4}} = \pm \frac{4x-3y+2}{\sqrt{16+9}}$

Now $a_1a_2 + b_1b_2 = (1)(4) + (-2)(-3) = 10$ i.e. > 0
hence for obtuse angle bisector take positive sign,
 $\Rightarrow (4 - \sqrt{5})x - (3 - 2\sqrt{5})y - (4\sqrt{5} - 2) = 0$
is the required obtuse angle bisector.

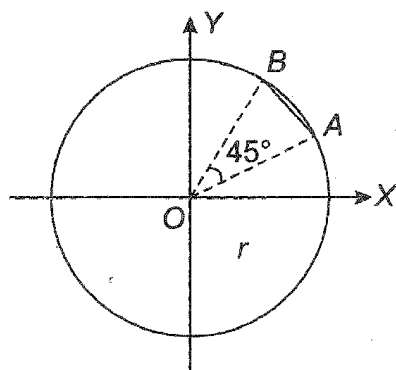
20. **Ans (A)**

Homogenise the equation of the circle with the help of the equation of the line

$$x^2 + y^2 = a^2 (1)^2$$

$$\Rightarrow x^2 + y^2 = a^2 (\ell x + my)^2$$

$$\Rightarrow (a^2 \ell^2 - 1)x^2 + 2\ell m a^2 xy + (a^2 m^2 - 1)y^2 = 0$$



$$\therefore \tan 45^\circ = \left| \frac{2\sqrt{\ell^2 m^2 a^4 - (a^2 y^2 - 1)(a^2 m^2 - 1)}}{a^2 \ell^2 - 1 + a^2 m^2 - 1} \right|$$

$$\text{Squaring, we get } 1 = \frac{4(a^2 \ell^2 + a^2 m^2 - 1)}{(a^2 + t^2 + a^2 m^2 - 2)^2}$$

$$\Rightarrow a^4(\ell^2 + m^2) - 8a^2(\ell^2 + m^2) + 8 = 0$$

$$\therefore \ell^2 + m^2 = \frac{8a^2 \pm \sqrt{64a^4 - 32a^4}}{2a^4}$$

$$a^2(\ell^2 + m^2) = 4 \pm 2\sqrt{2}$$

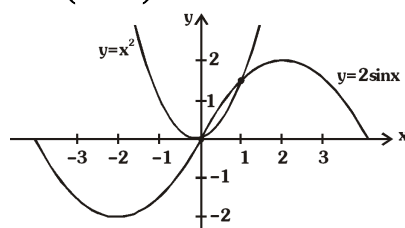
SECTION-II

1. **Ans (23.00)**

$$PA^2 + PC^2 = PB^2 + PD^2$$

$$\Rightarrow 4 + 100 = 81 + k \Rightarrow k = 23$$

2. **Ans (2.00)**



So total solutions are 2.

3. **Ans (11.00)**

Let point is $(\lambda, 4 - \lambda)$ as per equation,

$$\left| \frac{4\lambda + 3(4 - \lambda) - 10}{5} \right| = 1$$

$$\Rightarrow \lambda = 3, -7$$

4. **Ans (2.00)**

Point of intersection of the given lines are

$$\left(\frac{5}{3 + 4m}, \frac{3 + 9m}{3 + 4m} \right)$$

x-coordinate is an integer.

$$\therefore \frac{5}{3 + 4m} \text{ is an integer.}$$

$3 + 4m$ is divisor of 5

$$3 + 4m = 1, 3 + 4m = -1$$

$$3 + 4m = 5, 3 + 4m = -5$$

$$\Rightarrow m = -1, -2$$

5. **Ans (-4.00)**

$$(\cos 5\theta + \cos 2\theta) - (\cos 4\theta + \cos 3\theta)$$

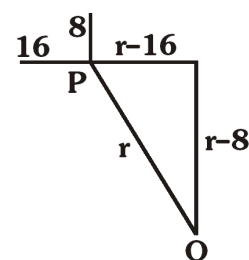
$$= 2 \cos \frac{7\theta}{2} \cos \frac{3\theta}{2} - 2 \cos \frac{7\theta}{2} \cos \frac{\theta}{2}$$

$$= 2 \cos \frac{7\theta}{2} \left[\cos \frac{3\theta}{2} - \cos \frac{\theta}{2} \right]$$

$$= -4 \sin \frac{\theta}{2} \sin \theta \cos \frac{7\theta}{2}$$

$$\therefore \lambda = -4$$

6. **Ans (40.00)**



Let $OP = r$ cms

$$r^2 = (r - 16)^2 + (r - 8)^2$$

$$\Rightarrow r = 8 \text{ or } 40$$

but for $r = 8$, $r - 16$ is negative

hence $r = 40$

7. **Ans (5.00)**

here we have $1 - 2 \sin^2 x + a \sin x = 2a - 7$

$$\Rightarrow 2 \sin^2 x - a \sin x + 2a - 8 = 0$$

$$\Rightarrow \sin x = \frac{a \pm \sqrt{a^2 - 8(2a - 8)}}{4}$$

$$\Rightarrow \sin x = \frac{a - 4}{2} \text{ or } 2 \text{ (reject)}$$

$$\Rightarrow -1 \leq \frac{a - 4}{2} \leq 1$$

$$\Rightarrow 2 \leq a - 4 \leq 2$$

$$\Rightarrow 2 \leq a \leq 6$$

$$\Rightarrow a = 2, 3, 4, 5, 6$$

8. **Ans (1.00)**

$$= 1 - \cos^2 12^\circ + 1 - \cos^2 21^\circ + \sin^2 39^\circ + \sin^2 48^\circ - \left(\frac{1 - \cos 18^\circ}{2} \right) - \left(\frac{1 - \cos 36^\circ}{2} \right)$$

$$= 2 - (\cos^2 12^\circ - \sin^2 48^\circ) - (\cos^2 21^\circ - \sin^2 39^\circ) - 1 + \frac{\cos 18^\circ}{2} + \frac{\cos 36^\circ}{2}$$

$$= 2 - (\cos 60^\circ \cos 36^\circ) - (\cos 60^\circ \cos 18^\circ) - 1 + \frac{\cos 18^\circ}{2} + \frac{\cos 36^\circ}{2}$$

$$= 2 - \frac{\cos 36^\circ}{2} - \frac{\cos 18^\circ}{2} - 1 + \frac{\cos 18^\circ}{2} + \frac{\cos 36^\circ}{2} = 1$$

9. **Ans (-19.00)**

$$* m \cdot m^2 = \frac{a}{1} \Rightarrow m^3 = a$$

$$* m + m^2 = 6$$

$$\Rightarrow m(1 + m) = 6 \text{ \{Taking cube on both side\}}$$

$$\Rightarrow m^3(1 + m^3 + 3m(1 + m)) = 216$$

$$\Rightarrow a[1 + a + 3 \times 6] = 216$$

$$\Rightarrow a^2 + 19a - 216 = 0$$

$$\Rightarrow (a + 27)(a - 8) = 0$$

$$\Rightarrow a = 8 \text{ or } -27$$

10. **Ans (468.75)**

As the point $(3\sqrt{3}, 2)$ lies on the line $y = \sqrt{3}x - 7$ so locus of P will be a line perpendicular to given line and passing through $(3\sqrt{3}, 2)$

$$y - 2 = -\frac{1}{\sqrt{3}}(x - 3\sqrt{3})$$

$$y - 2 = \frac{-x}{\sqrt{3}} + 3$$

$$\frac{x}{\sqrt{3}} + y - 5 = 0$$

It intersects x and y axis at $(5\sqrt{3}, 0)$ and $(0, 5)$ so

$$\text{area of the triangle formed} = \frac{5\sqrt{3} \times 5}{2} = \frac{25\sqrt{3}}{2} = A$$

$$\text{So } A^2 = 468.75$$